

Acute coronary Syndrome & Myocardial Infarction (IHD/ACS) Part 2

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives:

At the end of this session, students should be able to:

1. Relay the pertinent clinical features of acute myocardial infarction during a focused cardiac exam
2. Classify acute myocardial infarction based on clinical findings, ECG findings and laboratory findings
3. Explain the role of risk stratification in the prediction of the risk when managing patients with acute ischemic event
4. Demonstrate awareness of the complications of acute myocardial infarction

Session Outline:

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- Introduction to Acute Coronary Syndrome (ACS)
- Diagnosis and Clinical Presentation
- ECG Findings and Interpretation
- Cardiac Biomarkers
- Treatment and Chronic Management
- Complications

Introduction to Acute Coronary Syndrome (ACS)

What Is Acute Coronary Syndrome (ACS) ?

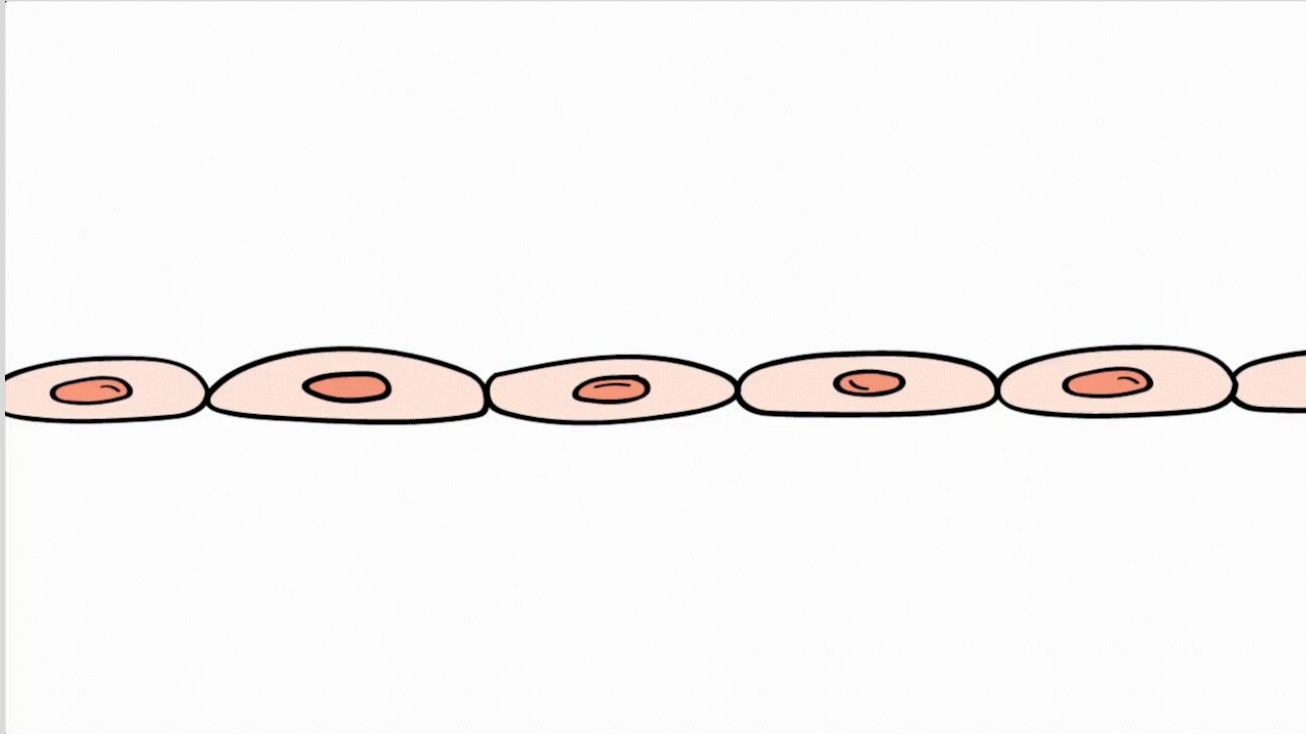
Acute Coronary Syndrome (ACS): An umbrella term used to describe a range of clinical conditions associated with a **sudden** and **severe decrease in blood flow** to the heart, causing symptoms such as **chest pain, shortness of breath**, etc.

- Unstable Angina
- Non-ST-Segment Elevation Myocardial Infarction (NSTEMI)
- ST-Segment Elevation Myocardial Infarction (STEMI)

Pathogenesis

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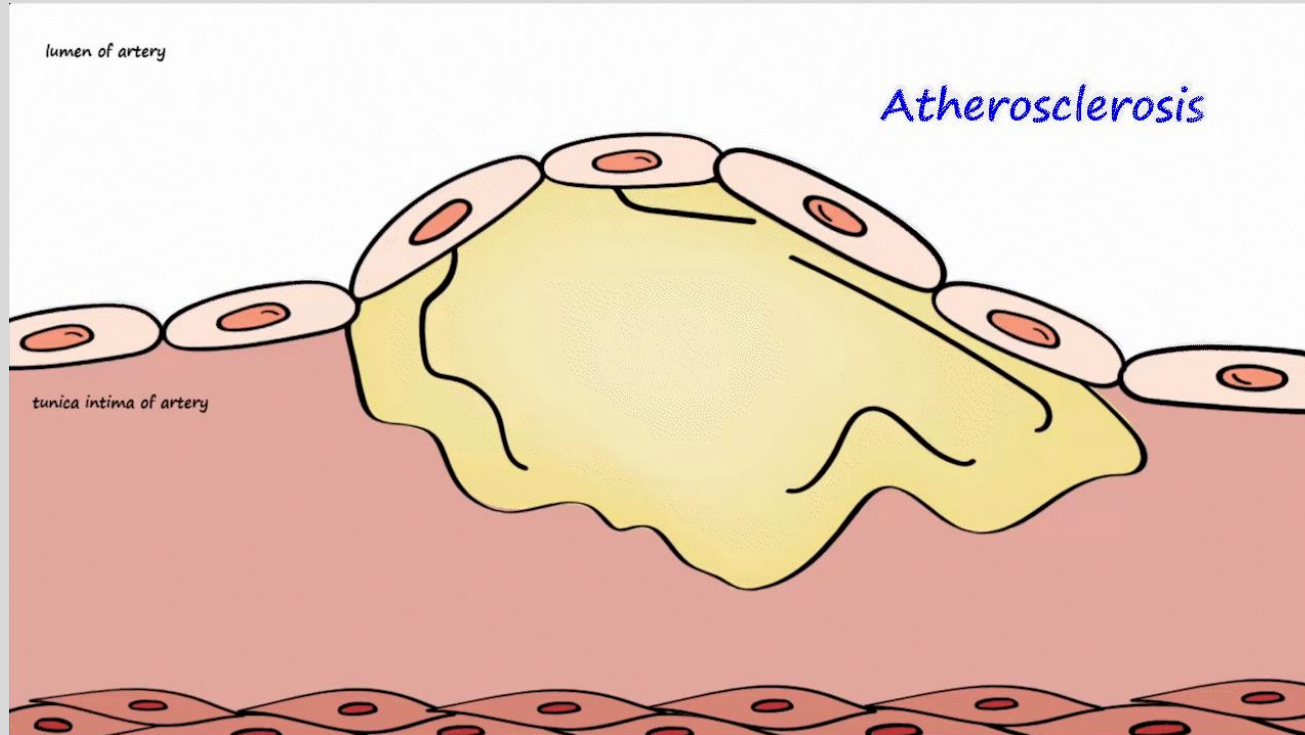
Source:

1. National Center for Health Statistics. Multiple Cause of Death 2018–2022 on CDC WONDER Database. Accessed May 3, 2024. <https://wonder.cdc.gov/mcd.html>
2. Martin SS, Aday AW, Almarzooq ZI, et al.; American Heart Association Council on Epidemiology and Prevention Statistics Committee; Stroke Statistics Subcommittee. [2024 heart disease and stroke statistics: a report of US and global data from the American Heart Association](#). *Circulation*. 2024;149:e347–913.
3. Tsao CW, Aday AW, Almarzooq ZI, et al. Heart Disease and Stroke Statistics—2023 Update: A Report From the American Heart Association. *Circulation*. 2023;147:e93–e621.
4. Gomez, J. (2017, October — exact date unavailable). *Pathogenesis of Atherosclerosis* [Video]. YouTube. <https://www.youtube.com/watch?v=N33Is8ezlPY>

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Our Patient:

Chief Complaint (CC): “Chest Pain”

History of Present Illness (HPI): 47-year-old black male that presents to the ED complaining of chest pain. He states that the pain is constant, and it radiates to his left shoulder. He describes the pain as a squeezing sensation in the middle of his chest. The pain began approximately 30 minutes ago and graded a 10/10. He was given sublingual nitroglycerin in the ambulance on the way to the ED, with no relief. PMHx is significant for hypertension and hyperlipidemia. His medications include 10mg oral amlodipine daily and 20mg oral rosuvastatin daily. On physical exam, the patient is tachycardiac, diaphoretic, and clenching his fists over his sternum.

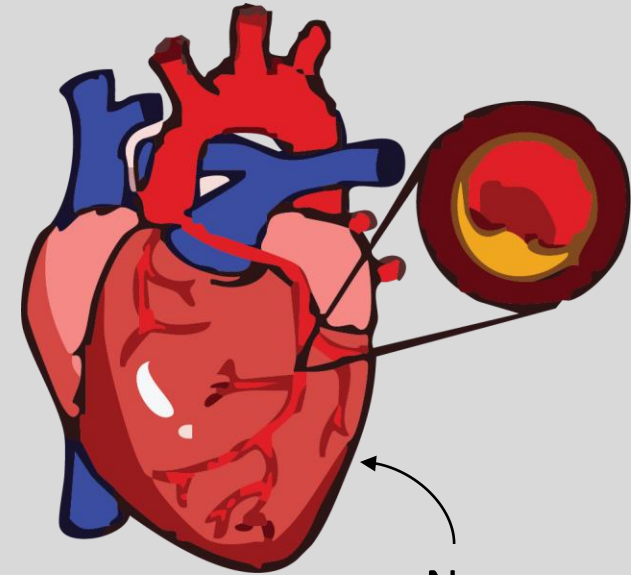
What is the suspected diagnosis?

Acute Coronary Syndrome (ACS)

- The term acute coronary syndrome (ACS) is applied to patients in whom there is a suspicion or confirmation of acute myocardial ischemia or infarction.
- Unstable angina, Non-ST-Elevation Myocardial Infarction (NSTEMI), and ST-Elevation MI (STEMI) are the three traditional types of ACS.
- Diagnosis is based on the Clinical Presentation, Serial ECGs and Cardiac Biomarkers

Unstable Angina

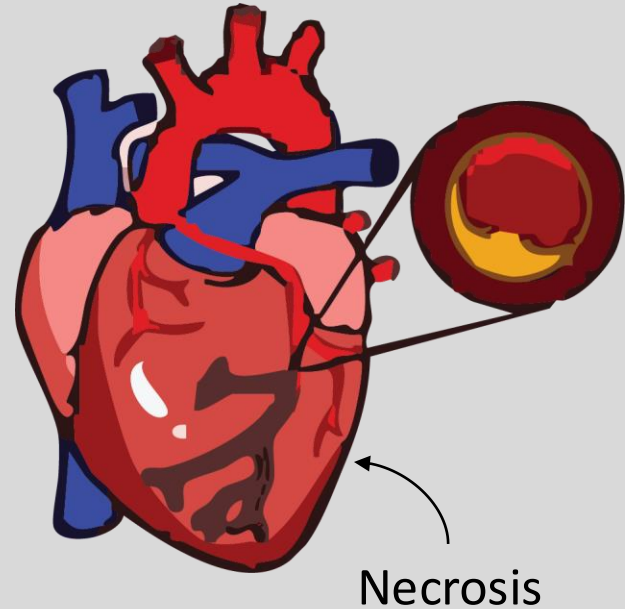
- **Partial/subtotal occlusion (~90%) of the coronary artery leading to subendocardial ischemia**
- **Acute, severe, substernal chest pain** with increasing frequency or intensity or duration
- Occurs **at rest** and **NOT relieved with SL nitroglycerin**
- **Lasts >10 minutes**
- +/- ST depressions and/or T-wave inversions (ECG changes indicative of ischemia)
- Normal troponin/CK-MB
 - **No cardiac biomarker elevation → No signs of myonecrosis**



No
Necrosis

Non-ST-Segment Elevation Myocardial Infarction (NSTEMI)

- **Partial/subtotal occlusion (~90%) of the coronary artery leading to subendocardial ischemia**
- **Acute, severe, substernal chest pain** with increasing frequency or intensity
 - Assoc. w/ radiation, diaphoresis, tachycardia
- Occurs **at rest** and **NOT relieved with SL nitroglycerin**
- **Lasts >10 minutes**
- +/- ST depressions and/or T-wave inversions (ECG changes indicative of ischemia)
- **+ Troponin, + CK-MB**
 - Cardiac biomarker elevation → Myonecrosis



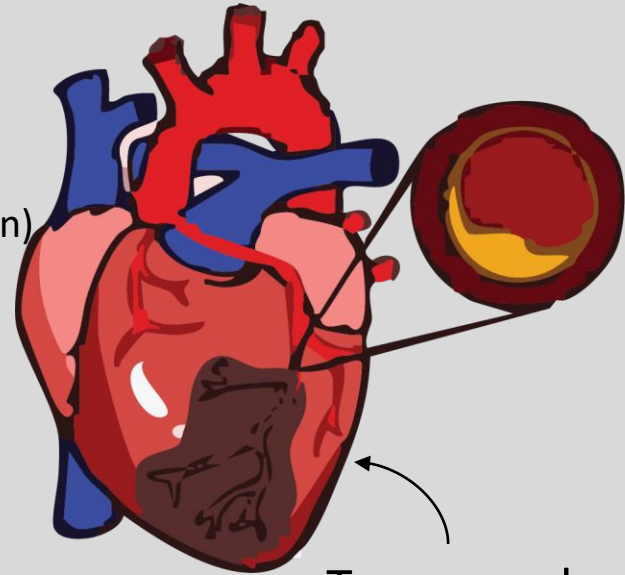
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2. Simons M, Albert JS. Acute coronary syndrome: terminology and classification. In: UpToDate. Post TW (Ed). UpToDate. Waltham, MA: published October 2024. Accessed June 27, 2025. <https://www.uptodate.com/contents/acute-coronary-syndrome-terminology-and-classification>

ST-Segment Elevation Myocardial Infarction (STEMI)

- **Total occlusion (100%) of the coronary artery leading to transmural ischemia**
- **Acute, severe, substernal chest pain** with increasing frequency or intensity
 - Assoc. w/ radiation, diaphoresis, tachycardia, shock (hypotension)
- Occurs **at rest** and **NOT relieved with SL nitroglycerin**
- **Lasts >10 minutes**
- **+ ST Elevation** and/or **new Left Bundle Branch Block (LBBB)**
- **+ Troponin, + CK-MB**
 - Cardiac biomarker elevation → Myonecrosis



Transmural
Necrosis

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Unstable Angina vs. NSTEMI vs. STEMI

	<u>Unstable Angina</u>	<u>NSTEMI</u>	<u>STEMI</u>
Pain Presentation	Crushing/squeezing chest pain with rest lasting >10min	Crushing/squeezing chest pain with rest lasting >10min	Crushing/squeezing chest pain with rest lasting >10min
Myocardial necrosis	No Myocardial Necrosis	Myocardial Necrosis	Myocardial Necrosis
ECG changes	No EKG changes/ST depression or T wave inversion	No EKG changes/ST depression or T wave inversion	ST Elevation or new LBBB
Cardiac Biomarkers (Troponin, CK-MB)	None	+Troponin, +CK-MB	+Troponin, +CK-MB
% Occlusion	~90% Occlusion	~90% Occlusion	100% Occlusion

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- A. Gastroesophageal Reflux Disease (GERD)
- B. Costochondritis
- C. Stable angina
- D. Panic Disorder
- E. Acute Coronary Syndrome (ACS)

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What would be the most appropriate next step in management of this patient?

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- A. Send for emergency cardiac catheterization
- B. Electrocardiogram and Cardiac biomarkers
- C. Cardiology consultation
- D. Chest X-ray
- E. Troponin levels

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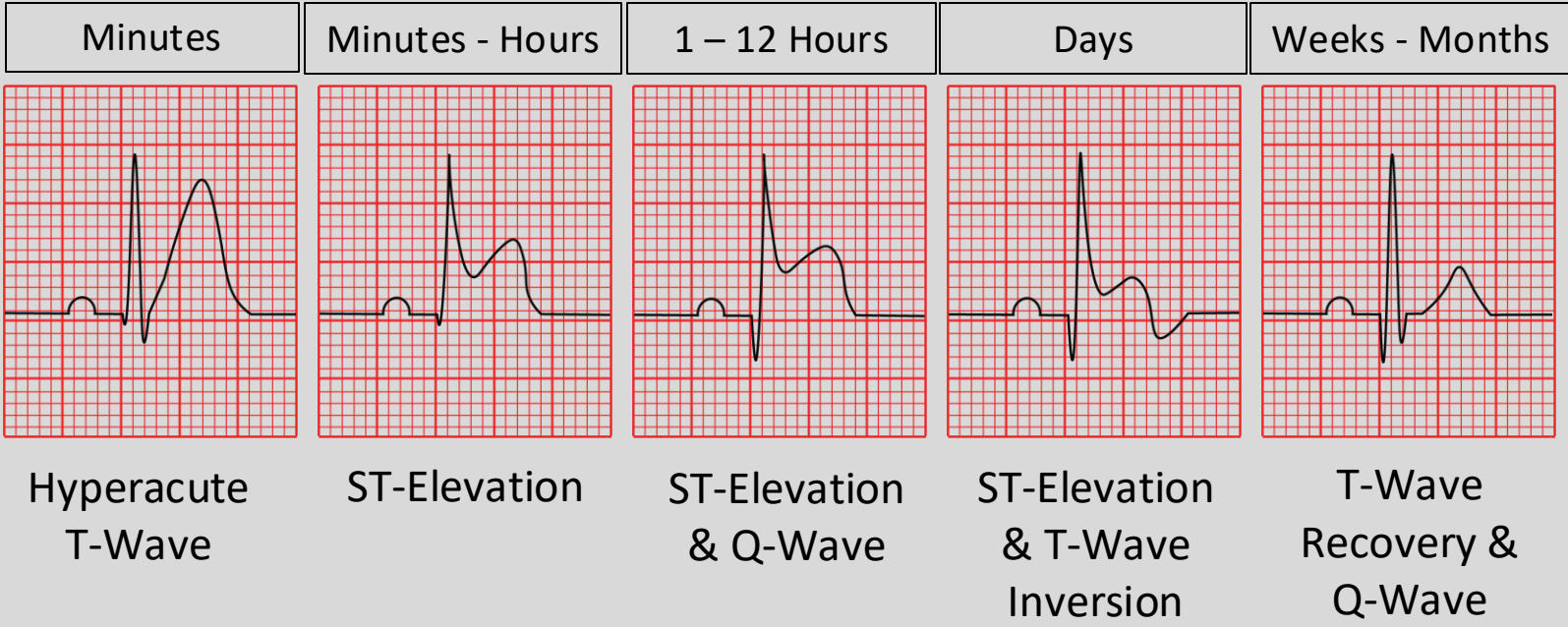
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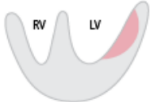
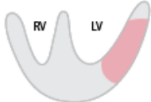
ECG Findings and Interpretation

STEMI ECG Progression



Source:
1. Thaler, M. S. (2022). The Only EKG Book You'll Ever Need (10th North American ed.). Lippincott Williams & Wilkins.

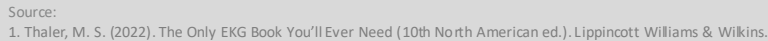
Summary

	Stable angina	Unstable angina	NSTEMI	STEMI
PAIN	On exertion	Mild exertion or at rest	At rest	At rest
TROPONIN LEVEL	No elevation	No elevation	Elevated	Elevated
INFARCTION	None	None	 Subendocardial	 Transmural
ECG CHANGES	None	Possible ST depression and/or T-wave inversion	ST depression and/or T-wave inversion	ST elevation
Myocardial infarction	Most often due to rupture of coronary artery atherosclerotic plaque → acute thrombosis. ↑ cardiac biomarkers (CK-MB, troponins) are diagnostic.			
	NSTEMI		STEMI	
INFARCT LOCATION	Subendocardial		Transmural	
LAYERS INVOLVED	Subendocardium (inner 1/3) especially vulnerable to ischemia		Full thickness of myocardial wall	
ECG CHANGES	ST-segment depression, T-wave inversion		ST-segment elevation, pathologic Q waves	

Source:
1. JLe, T., Bhushan, V., Qiu, C., Chalise, A., & Kaparaliotis, P. (2024). *First Aid for the USMLE Step 1* (34th ed.). McGraw-Hill Education.

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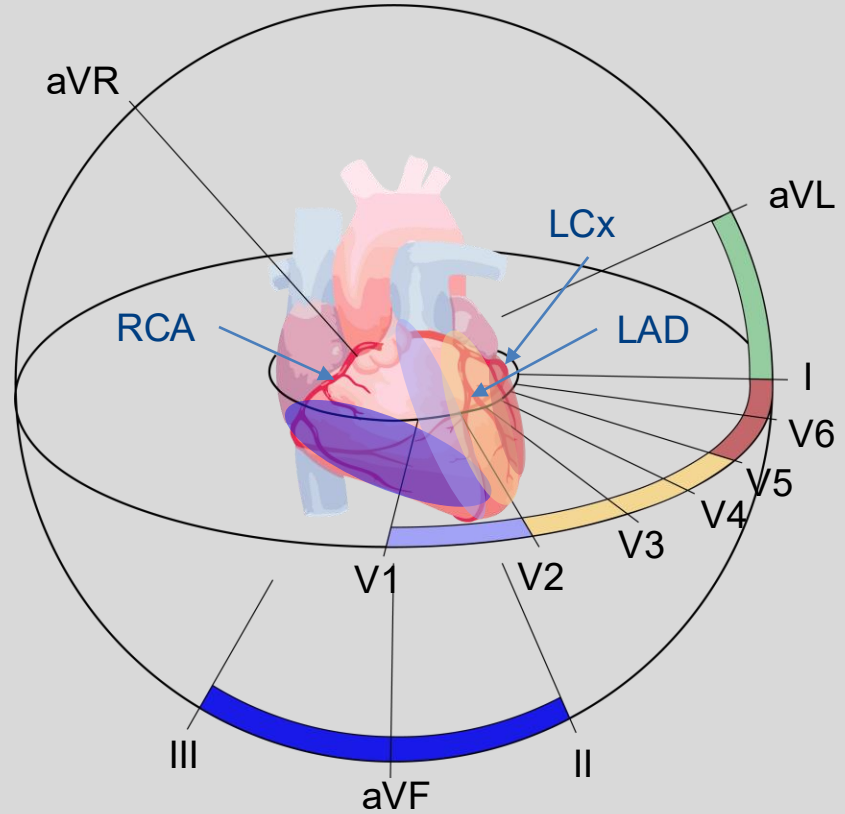
- Inferior Wall (Artery: RCA)
 - Leads II, III, aVF
- Septal (Artery: LAD)
 - Leads V1-V2
- Anterior Wall (Artery: LAD)
 - Leads V3-V4
- Anterolateral (Arteries: LAD, LCx)
 - Leads V5-V6
- Lateral Wall (Artery: LCx)
 - Lead I, aVL



Location of STEMI/NSTEMI

"I See All Leads"

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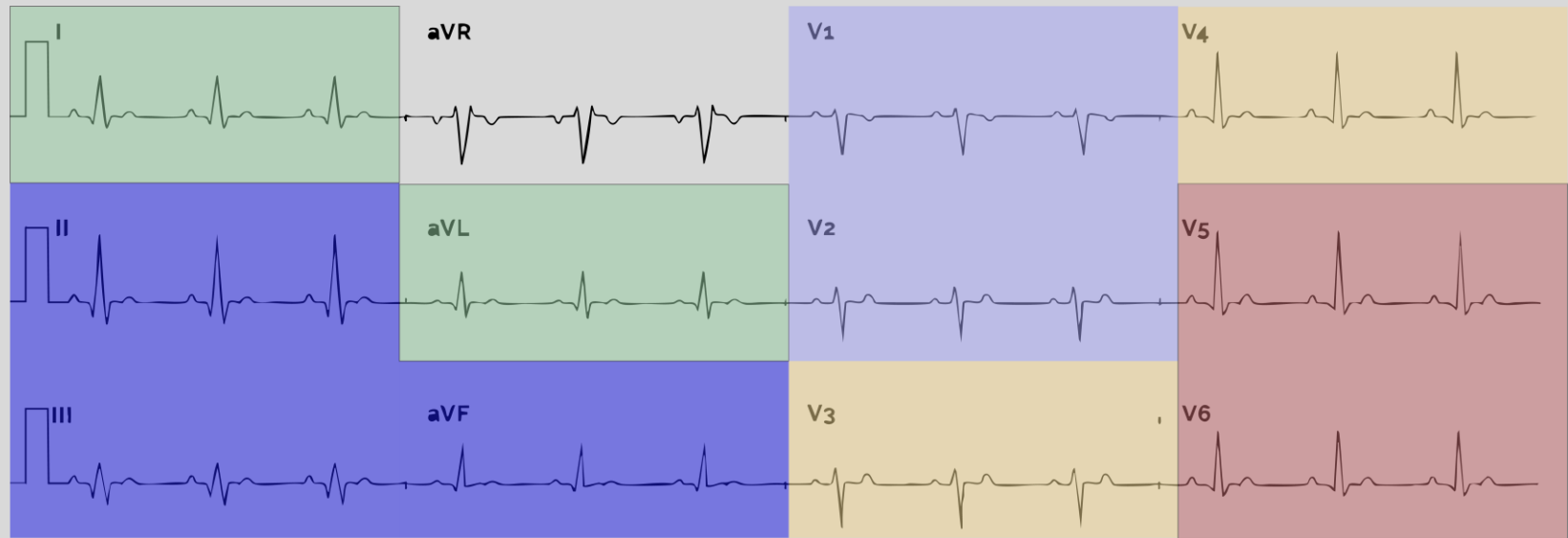
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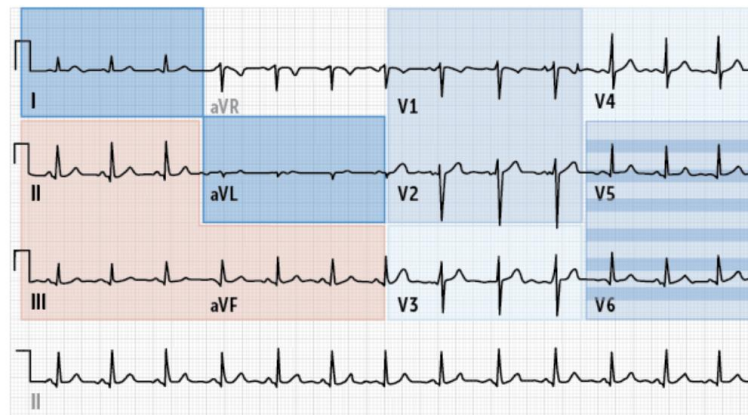
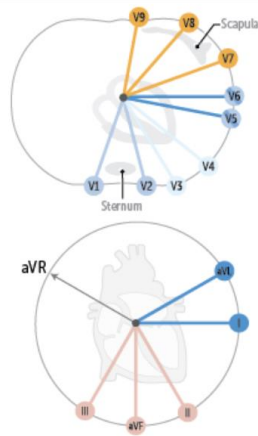


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Summary

ECG localization of STEMI

INFARCT LOCATION	LEADS WITH ST-SEGMENT ELEVATIONS OR Q WAVES
Anteroseptal (LAD)	V ₁ -V ₂
Anterapical (distal LAD)	V ₃ -V ₄
Anterolateral (LAD or LCX)	V ₅ -V ₆
Lateral (LCX)	I, aVL
InFerior (RCA)	II, III, aVF
Posterior (PDA)	V ₇ -V ₉ , ST depression in V ₁ -V ₃ with tall R waves

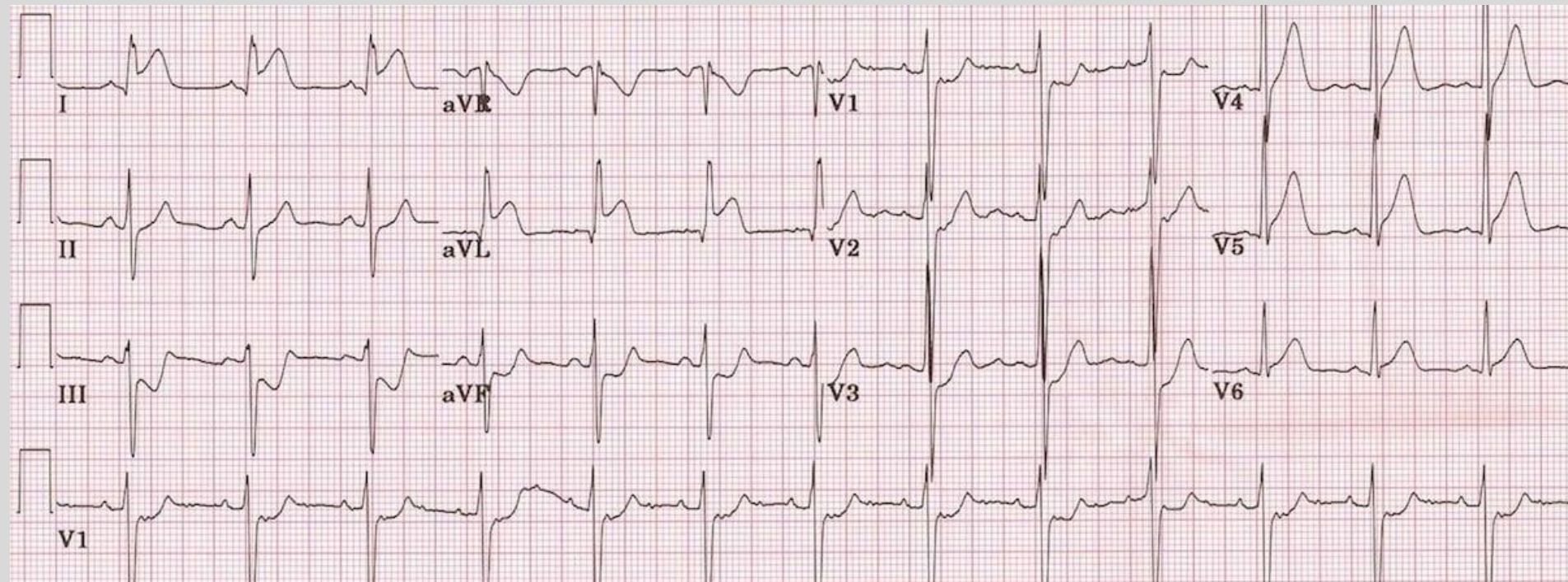


Our Patient:

He was admitted to the hospital. Labs and serial ECGs were performed.
His most recent ECG is shown below.

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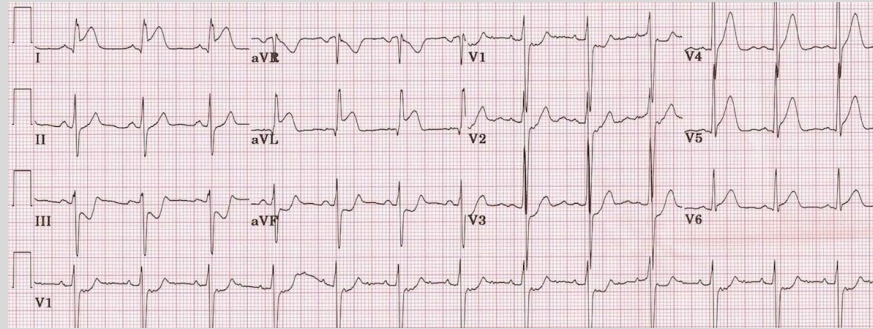
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Where is the myocardial infarction most likely located?

- A. Lateral Wall, due to LCx occlusion
- B. Inferior Wall, due to RCA occlusion
- C. Septal Wall, due to LAD occlusion
- D. Anterior Wall, due to LAD occlusion
- E. Anteriorlateral, due to LCx or LAD occlusion

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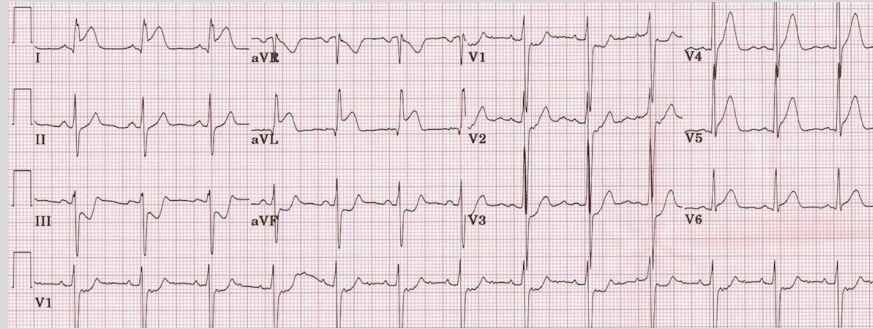
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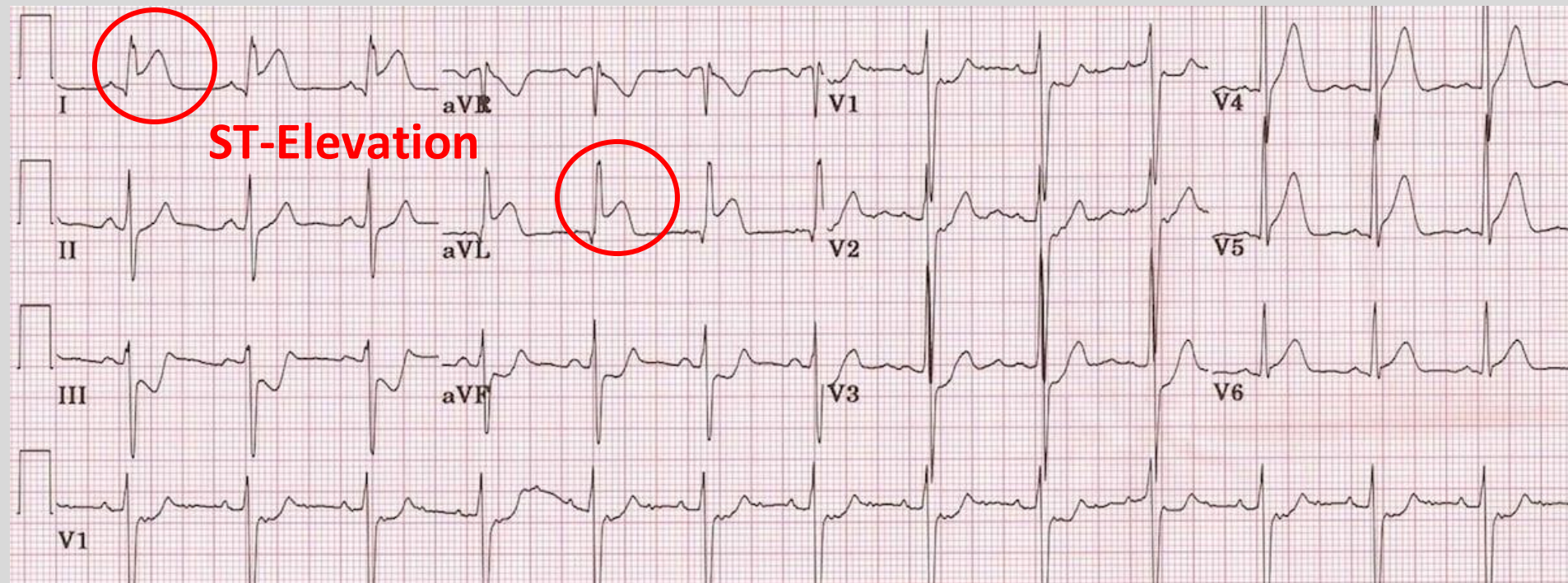
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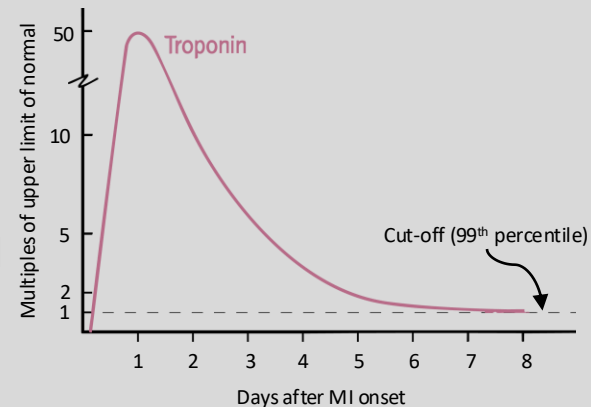
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Cardiac Biomarkers and Labs

Cardiac Biomarkers: Troponin

- Cardiac-Specific Troponin T (cTnT) & Cardiac-Specific Troponin I (cTnI)
- **Sensitive** (few false-negatives) **and Specific** (few false-positives)
 - Preferred biochemical markers for MI
 - In practical terms, the high-sensitivity troponin assays are of less immediate value in patients with STEMI, as urgent reperfusion strategies necessitate making a decision (based on a combination of clinical and ECG findings) before you obtain the blood results.
- **At least two sets are required 4 – 6 hours apart**
- First set should be drawn within 60 minutes from patient arrival
- Rises: **~ 4 hours**
- Peaks: 24 hours
- Return to Baseline: **7 – 10 days**

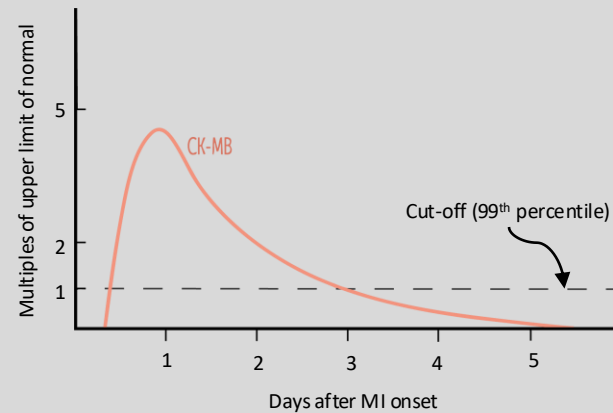


Cardiac Biomarkers: Creatine-Kinase

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- Creatine Phosphokinase (CK) & its MB isoenzyme (CK-MB)
- Less Sensitive and Specific than Troponins
- **Useful for looking at reinfarctions**
- Rises: **4 - 8 hours**
- Peaks: 16 - 24 hours
- Return to Baseline: **48 – 72 hours**



Source:

1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.

Labs: Other

- **Collect Routine Labs** - Evaluate for metabolic abnormalities, acute kidney injury, anemia, thrombocytopenia, and coagulopathy.
 - CBC, CMP, Kidney Function Studies, Coagulation Panel
- **Chest Radiograph** – Evaluate for other causes of chest discomfort.
- **Echocardiography, Computed Tomographic Angiography, and other imaging studies** – not routinely obtained unless a specific diagnosis is suspected
 - Aortic dissection
 - Pericardial tamponade

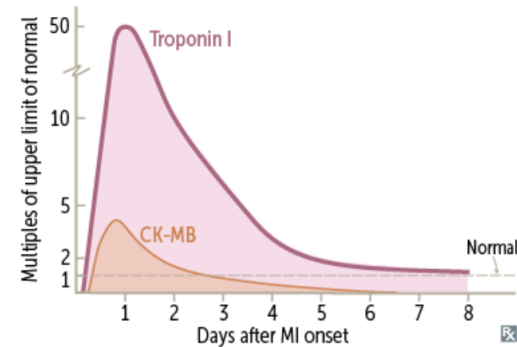
Summary

Diagnosis of myocardial infarction

In the first 6 hours, ECG is the gold standard. Cardiac troponin **I** rises after 4 hours (peaks at 24 hr) and is $\uparrow$ for 7–10 days; more specific than other protein markers.

CK-MB increases after 6–12 hours (peaks at 16–24 hr) and is predominantly found in myocardium but can also be released from skeletal muscle. Useful in diagnosing reinfarction following acute MI because levels return to normal after 48 hours.

ECG changes can include ST elevation (STEMI, transmural infarct), ST depression (NSTEMI, subendocardial infarct), hyperacute (peaked) T waves, T-wave inversion, and pathologic Q waves or poor R wave progression (evolving or old transmural infarct).



Treatment and Chronic Management

Goals of Therapy

- The primary goal of STEMI management is to reduce the risk of death and the extent of permanent cardiac injury associated with MI.
- Because therapy for patients with STEMI becomes less effective with each minute its delivery is delayed, we want to rapidly treat patients with STEMI before treatment becomes ineffective.

Risk Stratification: HEART Score

- HEART Score
 - History
 - EKG
 - Age
 - Risk Factors
 - Initial Troponin
- Any ED patient with chest pain that the physician deems appropriate for an ACS workup.
- Score >3 is a greater risk for ACS and requires further workup.

HEART Score for Major Cardiac Events

Predicts 6-week risk of major adverse cardiac events in patients with chest pain.

INSTRUCTIONS

Use in patients ≥21 years old presenting with symptoms suggestive of ACS. Do not use if new ST-segment elevation ≥1 mm or other new EKG changes, hypotension, life expectancy less than 1 year, or noncardiac medical/surgical/psychiatric illness determined by the provider to require admission.

History	Slightly suspicious	0
	Moderately suspicious	+1
	Highly suspicious	+2
EKG 1 point: No ST deviation but LBBB, LVH, repolarization changes (e.g. digoxin); 2 points: ST deviation not due to LBBB, LVH, or digoxin	Normal	0
	Non-specific repolarization disturbance	+1
	Significant ST deviation	+2
Age	<45 0 45-64 +1 ≥65 +2	
Risk factors Risk factors: HTN, hypercholesterolemia, DM, obesity (BMI >30 kg/m ²), smoking (current, or smoking cessation ≤3 mo), positive family history (parent or sibling with CVD before age 65); atherosclerotic disease: prior MI, PCI/CABG, CVA/TIA, or peripheral arterial disease	No known risk factors	0
	1-2 risk factors	+1
	≥3 risk factors or history of atherosclerotic disease	+2
Initial troponin Use local, regular sensitivity troponin assays and corresponding cutoffs	≤normal limit	0
	1-3× normal limit	+1
	>3× normal limit	+2

0 points

Low Score (0-3 points)

Risk of MACE of 0.9-1.7%.

Source:
1. Marszalek M, Smith J. HEART Score for Major Cardiac Events: When to Use. In: MDCalc Medical Calculator [Internet]; updated 2024. Accessed July 29, 2025. <https://www.mdcalc.com/calc/1752/heart-score-major-cardiac-events#when-to-use>.

Initial Management: Routine Medical Therapy

“MONA-BASH”

Morphine

Oxygen

Nitrates

Aspirin

Beta-blockers

ACE inhibitors

Statins

Heparin

Initial Management: Routine Medical Therapy

Current Use (as of 2024 AHA/ACC guidelines)

Morphine	For refractory pain/anxiety not relieved by nitrates. May worsen outcomes if overused (↓ preload = ↓ coronary perfusion).
Oxygen	Only if O ₂ sat <90%, respiratory distress, or high-risk features. Routine use in normoxic patients is not recommended.
Nitrates	Sublingual nitro first-line unless hypotensive, RV infarct, or PDE-5 use.
Aspirin	Give immediately (chewed 162–325 mg). Core therapy.
Beta-blockers	Within 24 hours if no contraindications (e.g., HF, bradycardia, hypotension, asthma).
ACE inhibitors	Start within 24–48 hours, especially in LV dysfunction, anterior MI, or diabetes.
Statins	High-intensity statins immediately for all STEMI/NSTEMI patients unless contraindicated.
Heparin	UFH, LMWH, or bivalirudin, depending on PCI or fibrinolysis strategy.

* All patients with **Acute Coronary Syndrome** should receive Dual Antiplatelet Therapy (DAPT) immediately upon arrival in the emergency room -> include aspirin + P_{2Y}₁₂ receptor inhibitor (*clopidogrel*)

Initial Management: Choosing and Initiating Reperfusion Therapy

- The primary goal of acute management is to rapidly restore blood flow to the occluded coronary artery.
- Initial management requires rapid selection and administration of reperfusion therapy.
- Patients with UA/NSTEMI with one of following characteristics should also receive prompt reperfusion therapy:
 - Cardiogenic shock
 - Severe Heart Failure
 - Recurrent or persistent angina despite intensive medical therapy
 - New or worsening mitral regurgitation or new ventricular septal defect
 - Sustained ventricular arrhythmia
- Reperfusion can be achieved via:
 - Percutaneous Coronary Intervention (PCI)
 - Fibrinolysis
 - **Fibrinolysis is not recommended in patients with NSTEMI**

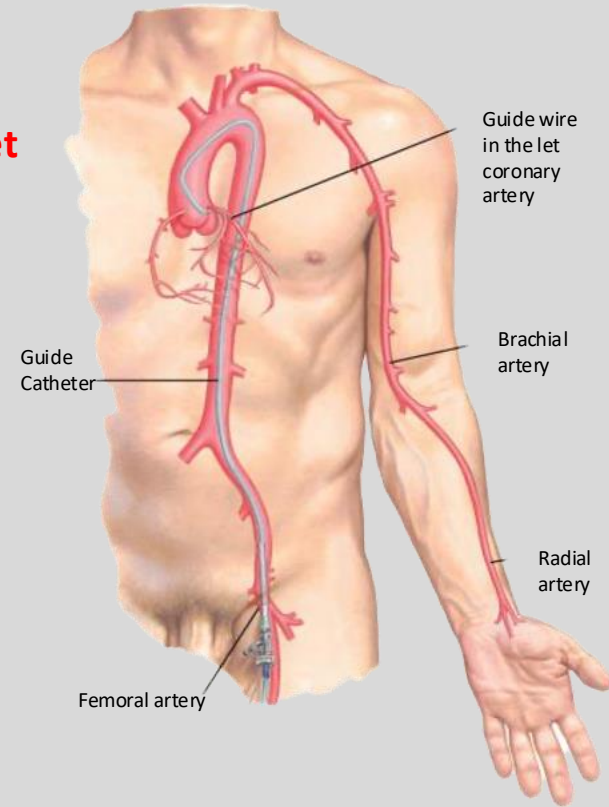
Reperfusion Therapy: Percutaneous Coronary Intervention (PCI)

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- All patients with STEMI who present within **12 hours of symptom onset** should undergo reperfusion.
- The approach to reperfusion depends on the availability of PCI.

* **“Door-to-balloon”** time goal for PCI = **<90 minutes**



Source:

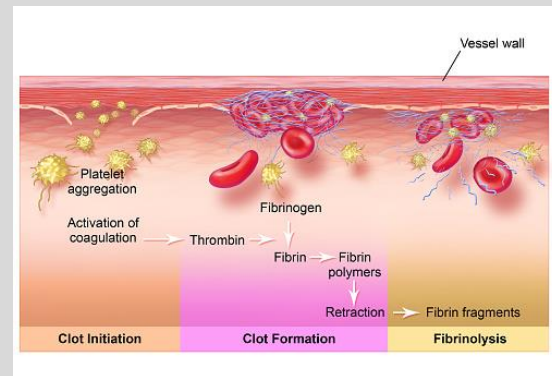
1. Reeder GS, Kennedy HL. Overview of the acute management of ST-elevation myocardial infarction. In: UpToDate, Wolters Kluwer; published approx. 3 months ago. Accessed July 28 2025.

Reperfusion Therapy: Fibrinolysis

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- In patients who present within **12 hours of symptom onset**, whom the **FMC-to-PCI interval is more than 120 minutes** AND can **safely receive fibrinolytic therapy**.
- Patients with STEMI who receive reperfusion with immediate fibrinolytic therapy **should receive fibrinolytic therapy within 30 minutes** of FMC
- Contraindications
 - **Previous intracranial hemorrhage (ICH)**
 - Known structural cerebral vascular lesion
 - **Known malignant intracranial neoplasm**
 - Ischemic stroke within three months
 - Suspected aortic dissection
 - Active bleeding or bleeding diathesis
 - Significant closed-head or facial trauma within three months
- **Not for treatment of NSTEMI**

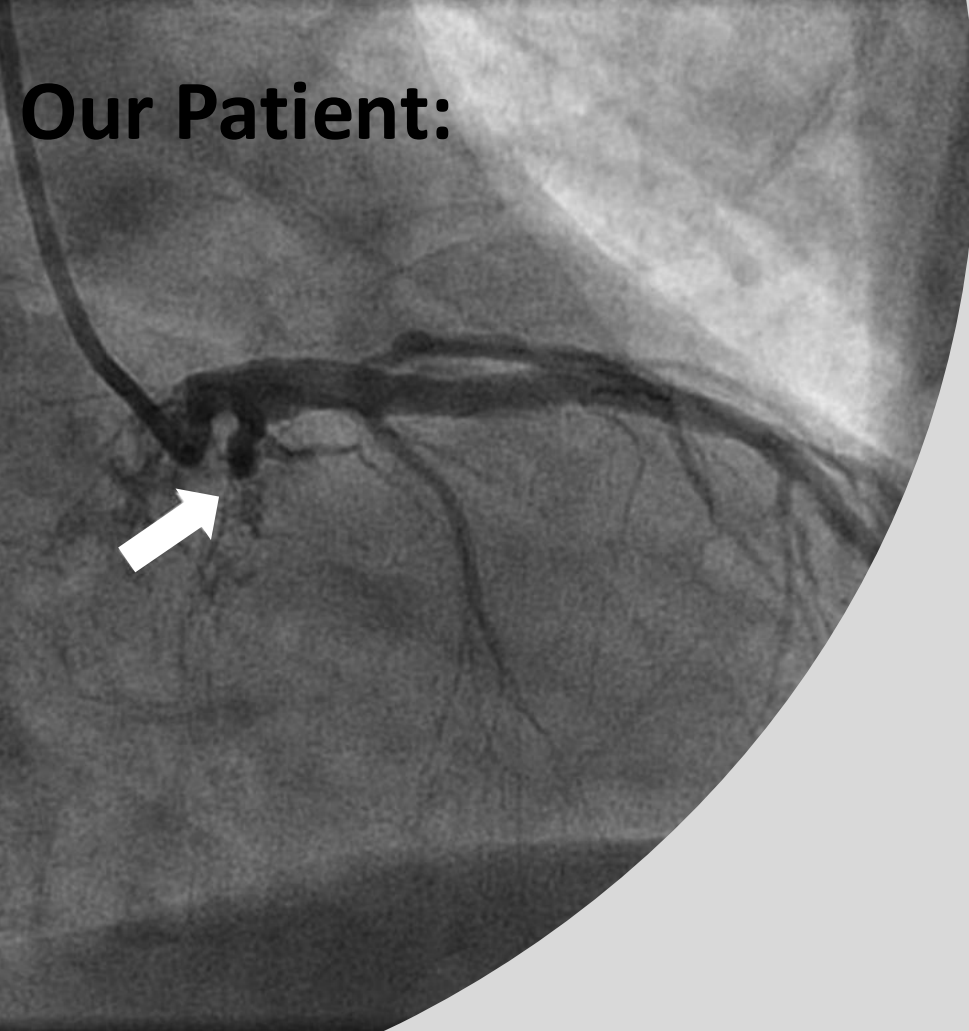


Source:

1. Reeder GS, Kennedy HL. Overview of the acute management of ST-elevation myocardial infarction. In: UpToDate, Wolters Kluwer; published approx. 3 months ago. Accessed July 28 2025.

What Happened to Our Patient?

Our Patient:

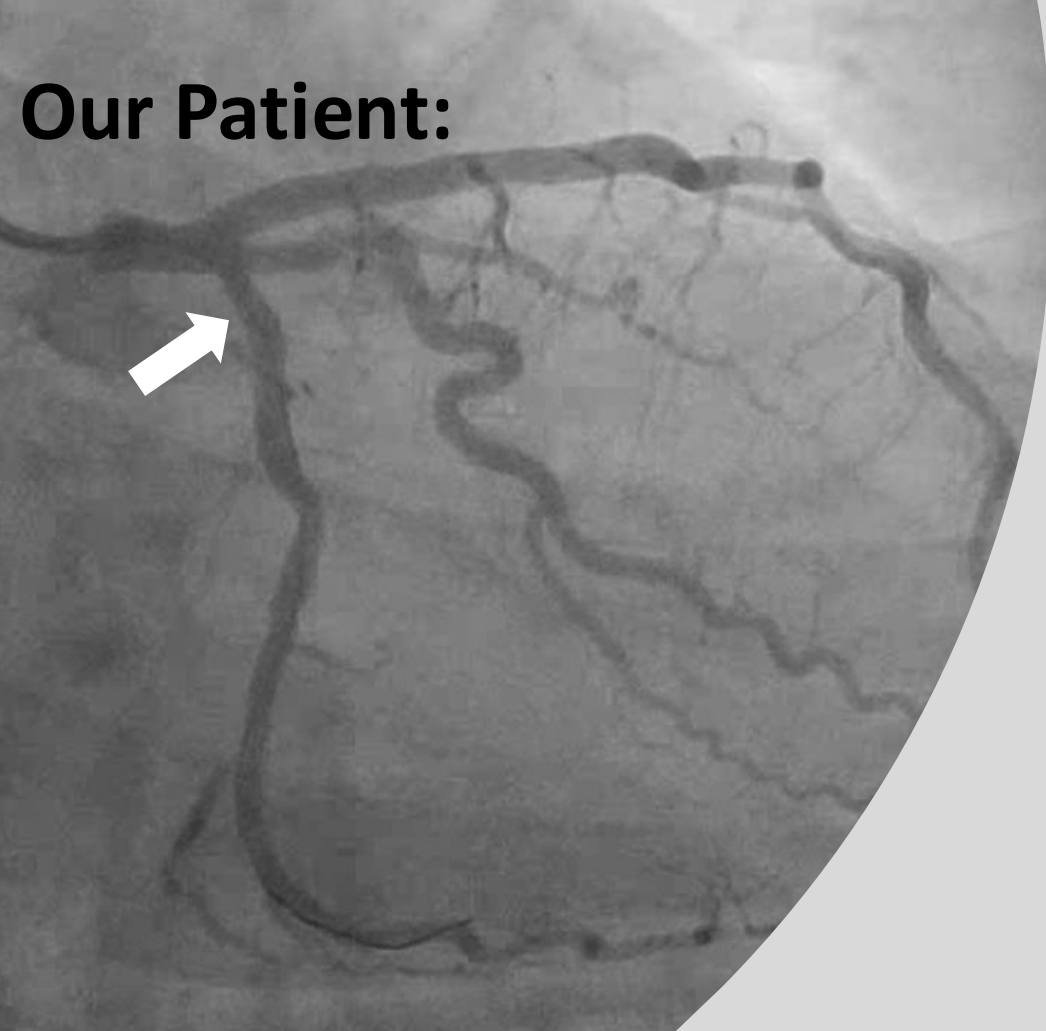


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The patient was rushed to Cath lab for immediate reperfusion of the LCX artery.

Our Patient:



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The patient was rushed to Cath lab for immediate reperfusion of the LCX artery.

Source:

1. <https://www.ctbsurgery.com/cor-angiogram-views.html>

Complications

Our Patient:

He was treated with PCI and a drug-eluting stent was placed in the left circumflex artery. After 1 week, the patient was discharged from the hospital and returned home on dual-antiplatelet therapy. Two months post discharge, the patient returns to the ED complaining of chest pain that worsens with inspiration and improves with leaning forward. On physical exam, a substernal friction rub is heard on auscultation. Labs show leukocytosis and elevated ESR and CRP. ON EKG, there is diffuse ST elevation in all the leads. Lungs are clear to auscultation b/l. There are no signs of pitting edema.

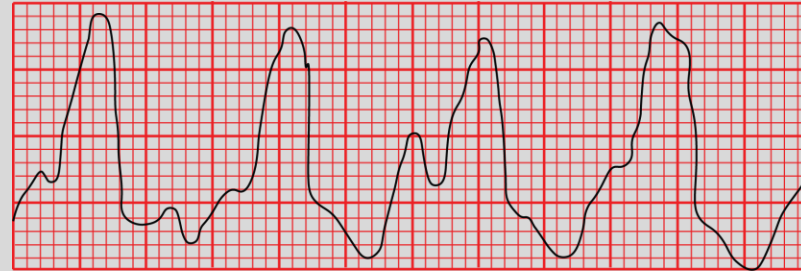
What is the most likely etiology of this patients chief complaint?

Post-MI Complications:

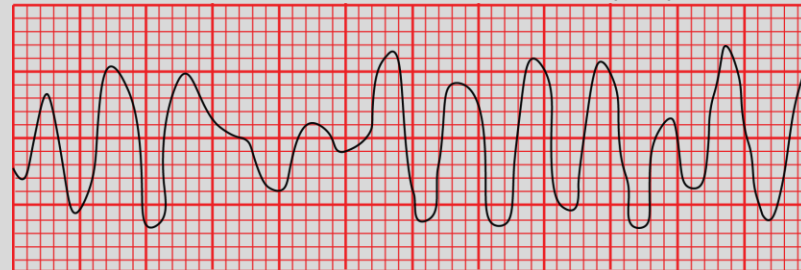
0 – 24 hrs

- **Ventricular Arrhythmias**
 - Most common early post-MI complication
 - Ventricular Tachycardia (VT) or Ventricular Fibrillation (VF)
 - Due to reentrant circuits in ischemic myocardium
 - Diagnosed by ECG

Ventricular Tachycardia (VT)



Ventricular Fibrillation (VF)



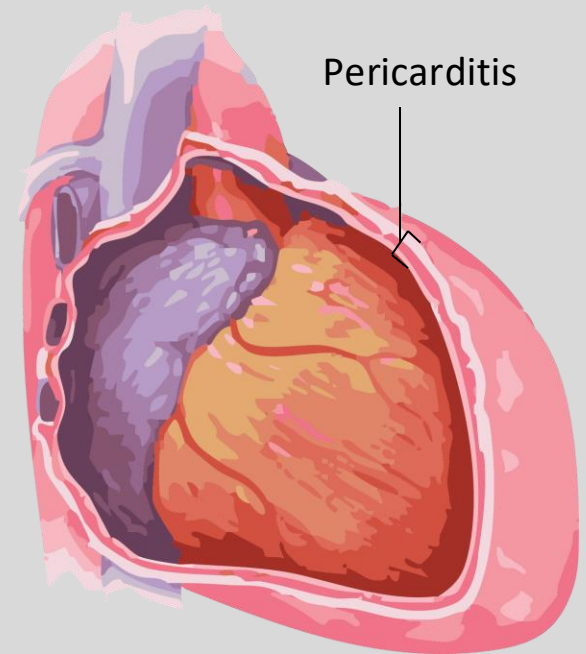
Source:

1. Jameson JL, Faud AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Zimetbaum, P. J., Amsdorf, M. F., & Josephson, M. E. (2006). *Conduction abnormalities after myocardial infarction*. In D. B. Rose (Ed.), *UpToDate* (14.2 ed.). Walters Kluwer.

Post-MI Complications:

1 – 3 days

- **Early Pericarditis**
 - Fibrinous pericarditis due to inflammation
 - Pleuritic chest pain, friction rub on exam
 - Often self-limiting; treated with NSAIDs if needed



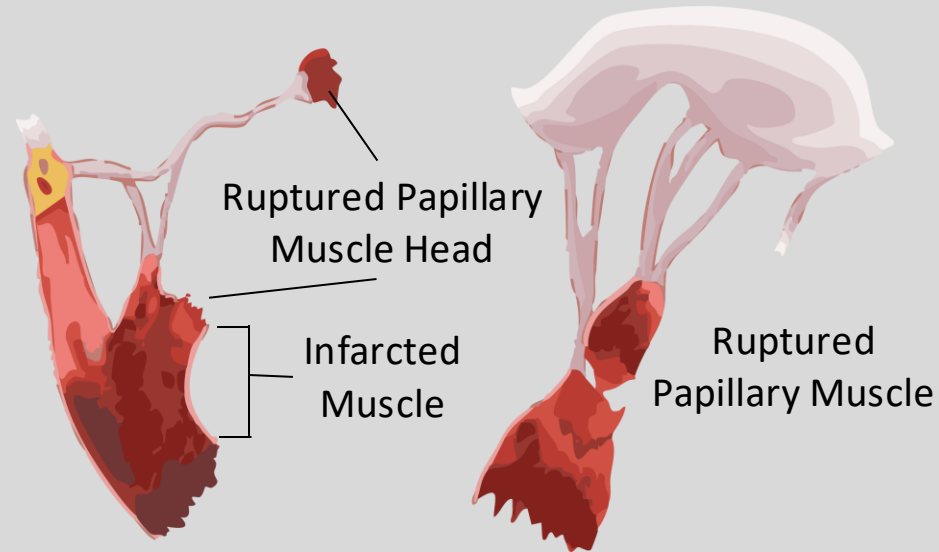
Source:

1. Jameson JL, Faud AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. LeWinter, M. M. (2024). *Pericardial complications of myocardial infarction*. In B. J. Gersh & J. K. Oh (Eds.), *UpToDate*. Retrieved July 30, 2025, from <https://www.uptodate.com/contents/pericardial-complications-of-myocardial-infarction>

Post-MI Complications:

3 – 5 days

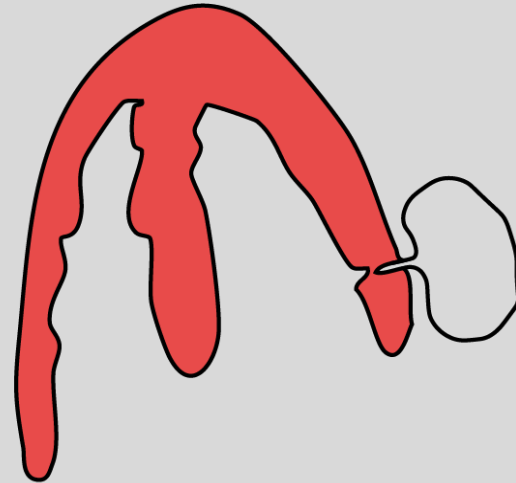
- **Papillary Muscle Rupture**
 - Typically affects posteromedial muscle (RCA infarct)
 - Leads to acute mitral regurgitation
 - Presents with new holosystolic murmur, pulmonary edema
- **Interventricular Septum Rupture**
 - Usually LAD infarct
 - Left-to-right shunt (VSD physiology)
 - Harsh systolic murmur, $\uparrow$ O₂ step-up in RV
 - Diagnosed by echo with Doppler



Post-MI Complications:

5 – 14 days

- **Ventricular Pseudoaneurysm**
 - Contained myocardial rupture
 - Risk of rupture remains high
 - Pulsatile mass, narrow neck on imaging
- **Ventricular Free Wall Rupture**
 - Catastrophic complication → cardiac tamponade
 - High risk in elderly, anterior MI, hypertension
 - Sudden death, pulseless electrical activity (PEA)



Pseudoaneurysm

Post-MI Complications:

2 weeks – months

- **True Ventricular Aneurysm**
 - Scarred myocardium bulges during systole
 - Persistent ST elevation on ECG
 - Wide neck, low rupture risk, may form thrombus
- **Post-cardiac Injury Syndrome (Dressler Syndrome)**
 - Autoimmune pericarditis
 - Fever, pleuritic chest pain, pericardial effusion
 - Treated with NSAIDs or colchicine



True Aneurysm

Summary

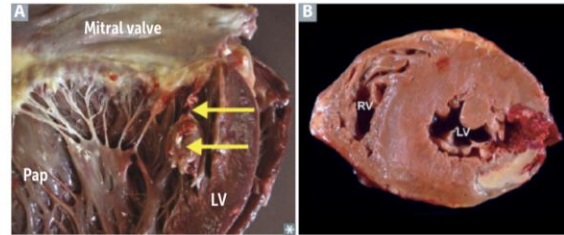
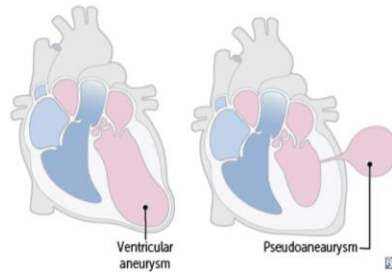
Myocardial infarction complications			
COMPLICATION	TIMEFRAME	FINDINGS	NOTES
Cardiac arrhythmia	First few days to several months	Can be supraventricular arrhythmias, ventricular arrhythmias, or conduction blocks.	Due to myocardial death and scarring. Important cause of death before reaching the hospital and within the first 48 hours post-MI.
Peri-infarction pericarditis	1–3 days	Pleuritic chest pain, pericardial friction rub, ECG changes, and/or small pericardial effusion.	Usually self-limited.
Papillary muscle rupture	2–7 days	Can result in acute mitral regurgitation → cardiogenic shock, severe pulmonary edema.	Posteromedial >> anterolateral papillary muscle rupture A , as the posteromedial has single artery blood supply (PDA) whereas anterolateral has dual (LAD, LCX).
Interventricular septal rupture	3–5 days	Symptoms can range from mild to severe with cardiogenic shock and pulmonary edema.	Macrophage-mediated degradation → VSD → ↑ O ₂ saturation and ↑ pressure in RV.
Ventricular pseudoaneurysm	3–14 days	May be asymptomatic. Symptoms may include chest pain, murmur, arrhythmia, syncope, HF, embolus from mural thrombus. Rupture → cardiac tamponade.	Free wall rupture contained by adherent pericardium or scar tissue — does not contain endocardium or myocardium. More likely to rupture than true aneurysm.
Ventricular free wall rupture	5–14 days	Free wall rupture B → cardiac tamponade; acute form usually leads to sudden death.	LV hypertrophy and previous MI protect against free wall rupture.

Source:
1. JLe, T., Bhushan, V., Qiu, C., Chalise, A., & Kaparaliotis, P. (2024). *First Aid for the USMLE Step 1* (34th ed.). McGraw-Hill Education.

Summary

Myocardial infarction complications

True ventricular aneurysm	2 weeks to several months	Similar to pseudoaneurysm.	Outward bulge with contraction ("dyskinesia"). Associated with fibrosis.
Postcardiac injury syndrome	Weeks to several months	Fibrinous pericarditis due to autoimmune reaction.	Also called Dressler syndrome. Cardiac antigens released after injury → deposition of immune complexes in pericardium → inflammation.



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- A. Thrombus formation occluding coronary vessel
- B. Arrhythmia
- C. Autoimmune reaction to the pericardium
- D. Decreased left ventricular ejection fraction
- E. Neutrophilic infiltration of the myocardium

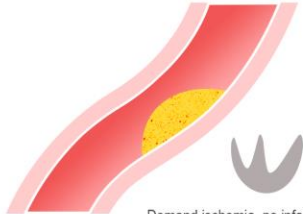
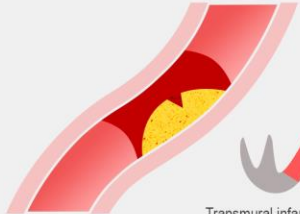
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- D. Decreased left ventricular ejection fraction
- E. Neutrophilic infiltration of the myocardium

Summary

ACUTE CORONARY SYNDROME	1 STABLE ANGINA Angina pain develops when there is increased demand in the setting of a stable atherosclerotic plaque. The vessel is unable to dilate enough to allow adequate blood flow to meet the myocardial demand.  Demand ischemia, no infarct	2 UNSTABLE ANGINA The plaque ruptures and a thrombus forms around the ruptured plaque, causing partial occlusion of the vessel. Angina pain occurs at rest or progresses rapidly over a short period of time.  Supply ischemia, no infarct	3 NSTEMI During an NSTEMI, the plaque rupture and thrombus formation causes partial occlusion to the vessel that results in injury and infarct to the subendocardial myocardium.  Subendocardial infarct	4 STEMI A STEMI is characterized by complete occlusion of the blood vessel lumen, resulting in transmural injury and infarct to the myocardium, which is reflected by ECG changes and a rise in troponins.  Transmural infarct	
	ECG	 Normal	 Normal, Inverted T waves, or ST depression	 Normal, Inverted T waves, or ST depression	 Hyperacute T waves or ST elevation
	TROPONINS	Normal	Normal	Elevated	Elevated

This infographic was created by Paula Sneath and Leah Zhao for the Sirens to Scrubs series of CanadiEM.org.



Questions?

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Lecture Feedback Form:

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